

Project Overview: Exploring Non-Linearity in Regression

Objective

This project investigates how different modeling techniques capture non-linearity in data, with a focus on polynomial regression, basis functions, and regression splines. The Boston housing dataset and synthetic functions are used to illustrate concepts.

Question 2: Exploring Function Complexity

- Visualizes how models change as flexibility increases:
 - $m = 0$: Constant function
 - $m = 1$: Linear function
 - $m = 2$: Quadratic function
 - $m = 3$: Cubic function
- Demonstrates interpolation when $\lambda = 0$ and $m = 3$, leading to overfitting.
- Highlights the bias-variance trade-off visually.

Question 3: Basis Function Regression

- Implements a piecewise-defined basis function.
- Uses conditional logic to fit different segments of the curve.
- Shows how linear regression can model non-linear shapes when enhanced with custom basis terms.
- Annotates key points and highlights the model's structure and smooth transition.

Question 9: NOX vs. DIS Modeling (Boston Housing Data)

- **Polynomial Regression:**
 - Fits polynomial models (degrees 1–10) to predict NOX from DIS.
 - Degree 3 (cubic) fit visualized clearly.
 - Residual Sum of Squares (RSS) decreases with degree but overfitting may occur.

- **Cross-Validation for Polynomials:**
 - Applies 10-fold CV to select the best polynomial degree.
 - CV error plotted to show the ideal trade-off point.
- **Regression Splines:**
 - Uses B-splines to flexibly model NOX with 4 degrees of freedom.
 - Explores spline fits for 3–10 degrees of freedom.
 - Compares RSS values across models.
- **Cross-Validation for Splines:**
 - Applies CV to determine the best degree of freedom.
 - Plots error curve to visualize optimal spline complexity.

Key Takeaways

- **Model Complexity:**
 - Increased flexibility (higher degree or df) improves fit but may overfit.
- **Polynomial Models:**
 - Powerful but sensitive to the degree chosen.
- **Splines:**
 - Offer smoother fits and better control over local structure.
- **Cross-Validation:**
 - Crucial for selecting model complexity that generalizes well.